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SUPERSEDING
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FEDERAL SPECIFICATION

PIPE, STEEL (SEAMLESS AND WELDED) (FOR ORDINARY USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers black (mill) and hot-dipped, zinc-coated, welded, and seamless steel pipe, as specified in ASTM A120.

1.2 Classification. Steel pipe covered by this specification shall be of the following weights and classes, as specified (see 6.3):

Weight A - Standard weight.
Weight B - Extra strong.
Weight C - Double extra strong.
Class 1 - Black or mill finish.
Class 2 - Zinc-coated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

A90 - Weight of Coating on Zinc-Coated (Galvanized) Iron or Steel Articles.
A120 - Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A242 - High Strength Low Alloy Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Pipe shall be plain carbon steel produced by any process that meets the requirements of ASTM A120. When specified (see 6.2), pipe shall be made only of low alloy steel conforming to ASTM A242. Pipe may be seamless, electric-resistance-welded, or butt-welded.

3.2 Lengths. Unless otherwise specified (see 6.2), pipe shall be furnished in random or double random lengths, as specified in ASTM A120.

3.3 Diameter and thickness. Pipe size shall be of the diameters and thicknesses specified in tables I, II, III, and IV of ASTM A120. Unless otherwise specified (see 6.2), the wall thickness for standard weight pipe shown in table II of ASTM A120, shall be as follows:

Pipe size, inches	Wall thickness, inches
8	0.322
10	0.365
12	0.375

3.4 Permissible variations in weight and dimensions. The permissible variations in weight, outside diameter, and wall thickness shall be as shown in ASTM A120.

3.5 Class 1, black or mill finish pipe. Black or mill finish pipe shall be furnished without interior coating, but may be painted any color on the outside surface by the manufacturer for identification. This shall not exclude the usual temporary rust preventive treatments applied either on the inside or outside at the time of manufacture.

3.6 Class 2, zinc-coated pipe. Pipe shall be galvanized to comply with ASTM A120 as to process, grade of zinc, and weight of coating.

3.6.1 Zinc coating. Zinc coating shall be applied by the hot-dip process, shall be bright in appearance, smooth, continuous, and substantially free from blisters, lumps, gritty areas, acid spots, dross, warts, flux, and excess zinc. The weight of the zinc coating shall conform to the requirements of ASTM A120. When specified (see 6.2), the molten zinc bath shall contain not more than 2 percent of elements other than zinc. The composition of the bath shall be determined from samples selected at a point located at half the depth of the bath.

3.6.2 Zinc-coated pipe. Zinc coating shall be applied to pipe, before threading.

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3.6.3 Unless otherwise specified (see 6.2), zinc-coated couplings coated to comply with ASTM A120 shall be furnished with class 2 pipe.

3.7 Pipe, ends and couplings.

3.7.1 Weight A pipe. Unless otherwise specified (see 6.2), weight A, standard pipe shall be taper-threaded at both ends and shall have a coupling screwed on one end. Couplings on sizes 2-1/2-inches and larger shall be taper-tapped. Couplings on sizes 2 inches and smaller may be straight-tapped. Couplings shall be screwed on handling tight.

3.7.2 Weights B and C pipe. Unless otherwise specified (see 6.2), weight B, extra strong pipe, and weight C, double extra strong pipe, shall be furnished with square-cut ends.

3.7.3 Unthreaded pipe. Unless otherwise specified (see 6.2), weight A, unthreaded pipe shall be beveled 30 to 40 degrees on the outside, with a root face 1/16-inch $\pm$ 1/32-inch at the end of the pipe. The angle of the bevel shall be measured from a line perpendicular to the axis of the pipe.

3.8 Pipe threads. Pipe threads shall be in accordance with NBS H28. Unless otherwise specified (see 6.2), threaded pipe sized 4 inches and larger shall have a thread protector on the end not protected by the coupling. When specified, pipe in sizes 3-1/2 inches and smaller shall be furnished with all threads adequately protected. The end not protected by the coupling shall be protected by a ring of suitable material such as a plastic cap, or as specified herein. In lieu of couplings ring protectors may be used for pipe having threads sized 2 inches or less. Pipe in sizes of 1-1/2 inches or smaller may be bundled with ends wrapped in burlap or other suitable material instead of being equipped with couplings or ring protectors.

3.9 Strength requirements. Each length of pipe shall withstand the internal hydrostatic pressure shown in tables I, II, III, and IV of ASTM A120 for the weight and size under inspection without resulting in leakage, bulges, or other defects when subjected to the test specified in 4.4.1.

3.10 Identification marking. Each length of pipe shall be legibly marked by rolling, stamping, or stenciling to show the name or brand of the manufacturer, ASTM A120, and the length. Small diameter pipe, may be bundled; and items of information marked on a tag securely attached to each bundle.

3.11 Treatment and painting. Unless otherwise specified (see 6.2), the pipe shall be treated and painted in accordance with the manufacturer's standard practice.

3.12 Workmanship. The finished pipe shall be commercially straight, and free of injurious defects. All pipe shall be cut at right angles to the axis of the pipe and all burrs shall be removed. Pipe shall be substantially free from rust. The zinc-coating shall be free of injurious defects or excessive roughness. Welded pipe shall have a sound weld along the entire length of the pipe. Butt-welded pipe shall be free of lamination, blisters, burns, and

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interior obstruction. Electric resistance-welded pipe shall be free of injurious electrode burns, objectionable rolled-in flash, and commercially free from protruding flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Warehouse procurement. When specified (see 6.2), pipe furnished from a supplier's warehouse (whether the supplier is manufacturer or a jobber) shall be accompanied by evidence submitted to show that the pipe has been manufactured in accordance with ASTM A120. Such evidence shall include the identification marking outlined in 3.10 and certified copies of the manufacturer's inspection records of the examination and tests made on the pipe being submitted for acceptance approval. The inspection requirements of 4.2 and 4.4 inclusive, together with all the inspection procedures (including examinations and tests) outlined in ASTM A120 shall apply only to the manufacturer, providing the manufacturer's inspection records clearly indicate compliance with the provisions of this specification. The requirements of 4.5 are the supplier's responsibility.

4.2 Inspection requirements. Inspection requirements shall conform to the manufacturer's standard practice when the manufacturer certifies that they comply with the requirements of ASTM A120.

4.2.1 Sampling for examination and test. Except as outlined in 4.2 and 4.3, sampling for examination shall conform to the manufacturer's standard pracitce. Sampling for tests shall be in accordance with 4.2 and 4.2.2.

4.2.2 Sampling for test of zinc-coating. One length of zinc-coated pipe shall be selected at random from each lot of 500 lengths or fraction thereof of each size. A piece shall be cut from each end of each sample length and subjected to the test of 4.4.2.1. The test pieces shall be approximately 4 inches long. For large diameter pipe, the test specimens may be sectioned along the axis of the pipe and such sectioned specimens shall include approximately 24 square inches of surface (inside plus outside).

4.3 Examination. Each of the sample lengths of pipe selected in accordance with 4.2.1 shall be rolled on suitable supports, its entire surface and the interior carefully examined for obstructions and other defects. Each sample length of pipe shall be measured or otherwise examined as necessaary to verify conformity with all requirements that do not involve tests. Pipe shall be as prescribed in ASTM A120.

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4.4 Test procedures.

4.4.1 Hydrostatic pressure test. Hydrostatic tests shall be made, when required, to show compliance with 3.9. Welded pipe, 2-inch size and larger, shall be jarred near one end while under pressure. Pipe showing any defects in this test shall be rejected. For seamless and electric-resistance welded pipe, the hydrostatic pressure shall be maintained for not less than 5 seconds.

4.4.2 Test for weight of zinc-coating. The weight of zinc-coating shall be determined by the method specified in sections 15 through 18 of ASTM A90. Both the minimum weight and the average coating weight of each sample length of pipe shall conform to the requirements of ASTM A120. If the result of the coating test for any lot does not conform to 3.6.1, retests shall be made with two additional lengths of pipe from the same lot. Both retests shall meet the requirements of 3.6.1 or the lot shall be rejected.

4.5 Inspection of preparation for delivery. The preserviation, packaging, packing, and marking of the pipe shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The pipe shall be preserved and packaged in accordance with the applicable requirements of MIL-STD-163.

5.1.2 Level C. The pipe shall be cleaned, preserved, and packaged in a manner that will afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. The supplier may rely on his own practice providing it fulfills these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A and B. The pipe shall be packed in accordance with the applicable requirements of MIL-STD-163.

5.2.2 Level C The pipe shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable at lowest rates to the carrier. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, the pipe shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable (see 6.2).

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6. NOTES

6.1 Intended use. Pipe covered by this specification is intended for ordinary use in steam, water, gas, and air lines, but not for close coiling, bending, flanging, or use in maximum sustained operating temperatures above 450° F. The pipe is also intended for use in mechanical service, such as handrails.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Weight and class of pipe required (see 1.2).

(c) When pipe shall be made of low alloy steel (see 3.1).

(d) Whether random or double random lengths are required, and when pipe shall be cut to specified lengths (see 3.2).

(e) When wall thickness may be less on 8-, 10-, and 12-inch pipe, or must be greater on 12-inch pipe (see 3.3).

(f) When samples of molten zinc must be selected at a point half the depth of the bath to show that impurities do not exceed 2 percent (see 3.6.1).

(g) When zinc-coated couplings need not be furnished with zinc-coated pipe (see 3.6.3).

(h) If weight A, unthreaded (plain) pipe is required (see 3.7.1).

(i) If coupling is not required for threaded pipe (see 3.7.1).

(j) If weight B or C pipe shall have beveled ends (see 3.7.2).

(k) If square cut ends are required for unthreaded weight A pipe (see 3.7.3).

(l) When pipe threads shall be protected in a different manner than that specified (see 3.8 and 6.5).

(m) When treatment and painting other than that specified is required (see 3.11).

(n) If certification evidence is required for warehouse procurement (see 4.1.1).

(o) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

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6.3 Length. Pipe should be ordered by the number of feet required. When pipe is furnished with couplings, the length includes the coupling screwed on the pipe wrench-tight (handling tight), as specified in 3.7, but does not include the thread protector.

6.4 Purchaser options. Ordinarily most of the options listed in 6.2 can be readily met by suppliers, but some options may require shipment directly from the mill and additional time for delivery. In addition to some of the options in 6.2, those of 6.5, 6.6 and 6.7 may also require direct mill shipments.

6.5 Thread protection. Pipe covered by this specification in sizes 3-1/2 inches and smaller is regularly furnished without thread protection at the end not protected by the coupling. The experience of some Government users is that under some conditions of handling such unprotected threads are subject to excessive damage. This specification contemplates that pipe will be handled with due consideration and that damage to unprotected threads will be avoided. Where rough handling in shipment or on the job cannot be avoided, adequate protection for all threads should be specified in the contract or order, or pipe with plain ends specified. Where protection is required for threads, direct mill shipment of the pipe may be necessary.

6.6 Processes. Pipe is to be furnished either as welded or seamless. Welded pipe may be obtained in the following sizes: 1/2 inch and larger for electric-resistance welded pipe, and 1/8 inch to 4 inches, inclusive, for butt-welded pipe. Butt-welded double extra strong pipe is manufactured in sizes 2-1/2 inches and smaller. Seamless pipe can be obtained in sizes 1/8 inch and over. Seamless pipe in smaller sizes may be cold drawn. When pipe must be made by a particular process (seamless, electric-resistance welded, or butt-welded), the process should be specified.

6.7 Couplings. On sizes 8 inch nominal and larger in weight A, and all sizes in weight B, if threads and couplings are required, pipe is to be furnished with recessed couplings. Standard weight steel pipe in sizes 2 inches and smaller can be specified without couplings. Couplings are not required to be of the same material as the pipe.

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